



An Effective Anomalous Sound Detection Method Based on Global and Local Attribute Mining

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Abstract

Recent anomalous sound detection (ASD) methods have achieved impressive performance by leveraging attribute information. However, domain shift and imbalance issues caused by inconsistent attributes between source and target domains remain a key challenge. In this paper, we propose a novel **global and local attribute mining** method for ASD (GLAM-ASD) with a Transformer based encoder consisting of teacher and student branches to exploit semantic-rich information. The student mines contextual information via masked patch reconstruction while the teacher obtains weights from the student via exponential moving average. Global and local attributes are then derived in an unsupervised way. The student branch is further optimized by global self-distillation and pseudo-label prediction via learned local attributes. Evaluations on the DCASE2023 Task2 benchmark demonstrate the superiority of GLAM-ASD, especially in its target domain generalization performance.

Index Terms: Anomalous Sound Detection, Self-supervised learning, Local Attributes, Global Attributes

1. Introduction

Anomalous Sound Detection (ASD) refers to the analysis of machine audio to determine whether the machine is operating normally. This has recently begun to be widely applied in industrial fields. DCASE ASD challenges [1, 2, 3] mirror real-world deployment scenarios by only providing normal sounds for training, and requiring models to learn how to distinguish anomalous sounds using only normal data. This task is particularly challenging due to the scarcity and diversity of anomalous sounds, as well as domain shifts caused by varying background noise, machine types and operating conditions.

Effective modeling approaches often rely on discriminative learning methods, such as ResNet [4, 5, 6], MobileNetV2 [7], and STgram-MFN [8]. These methods extract labels from training meta information such as machine IDs and attributes, treating different categories as anomalies and subsequently training models using an Outlier Exposure (OE) approach [9]. However, there is always a trade-off between accuracy and generalization when converting attributes into labels. When the label granularity is too fine, the model tends to overfit, whereas overly coarse labels may lead to the model overlooking differences between distinct attributes. To address this issue, Zhang et al. [10] proposed a multi-task learning framework that incorporates labels of different granularities, while Wang et al. [11] trained independent classifiers for each attribute, leveraging attribute diversity to enhance their generalization capability.

Existing methods generally tune hyperparameters for each machine type, which is possible where development and evaluation datasets share the same machine types as in past ASD

tasks [1, 2]. However, it may not be possible in some practical deployments to collect both normal and anomalous data. Detection then becomes a “first-shot” problem and the challenge is to adapt to a completely new machine type for which only normal data is available, and still be capable of accurately detecting anomalies. This is the focus of DCASE2023 Task 2 [3].

Generative methods such as autoencoder (AE)-based systems [12, 13, 14] are trained to reconstruct frame-level features of normal audio, which do not rely on labels. These methods operate under the assumption that anomalous audio will exhibit notably higher reconstruction errors. Generative methods are known for their strong generalization capabilities, which enable them to handle first-shot problems effectively. However, they may overgeneralize by incorporating certain anomalies that are well-reconstructed, leading to misclassification. Zavrtnik et al. [15] took a different approach. Highlighting the importance of local features in anomaly detection, they distinguished between normal and anomalous samples by simulating anomalies in local frame-level features. AnoPatch [16] was proposed to leverage a Transformer backbone pre-trained on the extensive AudioSet dataset and fine-tune it on task-specific machine audio. It was shown that modeling from patch-level features can effectively address the sparsity of machine audio. However, fine-tuning pre-trained complex models still requires substantial computational resources, which can result in inefficient inference and high memory consumption. In [4, 17], self-supervised representation learning methods were presented, which mainly rely on data augmentations like mixup [18] and FeatEx [4].

In this paper, we propose a novel **global and local attribute mining** based ASD (GLAM-ASD) method to address the first-shot problem, by exploiting a self-supervised method in both clip- and frame-level embedding spaces. As shown in Fig. 1, a Transformer based encoder consisting of student and teacher branches is designed to learn both attributes in an unsupervised way. Local attributes refer to specific patch-level patterns in machine audio, which may detect anomalies with fine granularity. Specifically, Mel-spectrograms are extracted from the raw audio input, then patchified and embedded into a patch sequence as described in [19]. Following [20, 21], the sequence is multi-cropped and masked, which is used as input to the student for performing masked patch reconstruction by exploiting the contextual information among visible patches. The teacher is then obtained from the student via exponential moving average (EMA), from which local attributes are learned by an online clustering algorithm on the whole input sequence.

Global attributes refer to clip-level embeddings that aggregate local attribute mappings and can detect anomalies with coarse granularity. Following multi-crop training in [20], the global attributes from the student branch reflect various combinations of local attributes, while those from the teacher branch

Table 1: Result on the DCASE2023 Task2 development and evaluation datasets, Ω denotes the official score.

Methods	Development								Evaluation							
	ToyCar	ToyTrain	bearing	fan	gearbox	slider	valve	Ω	ToyDrone	ToyNscale	ToyTank	Vacuum	bandsaw	grinder	shaker	Ω
AE [22]	52.83	48.23	56.71	59.90	64.60	68.46	52.83	56.91	56.20	50.78	56.48	78.25	64.76	59.97	68.63	61.05
No. 1 [23]	54.58	54.20	64.21	75.95	76.92	92.06	85.67	69.28	55.71	84.92	60.72	92.27	60.97	65.18	63.50	66.97
No. 2 [24]	58.92	55.92	62.62	66.66	77.77	83.96	68.61	66.56	52.89	71.90	70.73	91.48	55.47	64.76	70.98	66.39
SSL4ASD [17]	-	-	-	-	-	-	-	66.92	-	-	-	-	-	-	-	68.13
AnoPatch [16]	58.78	67.16	70.43	66.65	58.67	81.88	53.73	64.24	64.49	85.04	72.60	92.24	69.71	64.06	80.25	74.23
GLAM (w/o fine-tune)	57.11	48.02	62.82	58.45	65.74	92.77	68.91	62.57	58.51	83.26	69.37	88.31	65.11	64.52	75.77	70.78
GLAM (w/ fine-tune)	58.39	55.61	69.97	61.01	68.24	94.03	75.20	67.06	67.89	90.25	76.14	94.80	66.12	71.89	79.20	76.79

where τ_l is the temperature parameter, and $\langle \cdot, \cdot \rangle$ represents the cosine similarity. We treat c_{y_n} as the positive sample for the masked patch recovery $\hat{z}_{s,n}$ and the other local attributes in the codebook as negative samples. This encourages the student branch to learn discriminative local features and capture semantic relationships between them. Unlike traditional methods that compute MSE loss between recovery results and the original spectrogram, our method employs online clustering and contrastive learning in the deep feature space, enhancing the effectiveness and discriminability of the learned features, thereby enabling fine-grained anomalous audio detection.

2.1.2. Global Attributes Mining

We adopt a dual-branch structure inspired by DINO [20], where the student branch learns from the teacher branch via knowledge distillation, and the teacher branch is updated using the EMA of the student branch. We apply random cropping and various data augmentation strategies to the input audio to obtain different views. Longer audio clips are treated as the global view X^g in DINO, while shorter audio clips are treated as the local view X^l . All audio clips are fed into the student branch, but only the global view is fed into the teacher branch. During pre-training, the global audio representations are learned by matching the distributions between the global view and the local view.

In contrast to DINO, we replace the projection head with a codebook C to obtain the local attribute distribution of the audio. This local distribution is then pooled and the first-order statistics are adopted as the clip-level distribution of the audio regarding its local attributes, as shown below:

$$g(x) = \sum_{i=1}^N C f(x_i) \quad (4)$$

where $f(\cdot)$ represents the encoder of either the teacher or student branch, and $f(x_i) \in \mathbb{R}^D$. We then optimize the KL-divergence loss L_{KL} to align the output distribution of the student branch with the teacher branch's distribution $g_t(x)$, which is considered as the global attributes:

$$L_{KL} = \sum_{x \in X^g} \sum_{x' \in X^g \cup X^l, x' \neq x} H(P_t(x), P_s(x')) \quad (5)$$

where P_t and P_s represent the probability distributions of the teacher and student branches' outputs, and H is the cross-entropy loss function. To prevent model collapse, we apply centering and sharpening operations on the probability distributions $g(x)$, as follows:

$$P_t(x) = \text{softmax}((g_t(x) - c)/\tau_t) \quad (6)$$

$$P_s(x) = \text{softmax}(g_s(x)/\tau_s) \quad (7)$$

where c is the centre of the teacher distribution, and τ_s and τ_t are temperature parameters for sharpening.

2.2. Pre-training with global and local attributes

By combining global self-distillation learning with local attribute prediction, the GLAM loss can be computed as $L_{GLAM} = L_{LAP} + \lambda_1 L_{KL} + \lambda_2 L_{VQ}$, where λ_1 and λ_2 are hyperparameters that control the balance between global and local contributions, as well as the update of the codebook.

L_{LAP} and L_{VQ} learn discriminative features in the local feature space, creating a high-quality codebook as local attributes. L_{KL} utilizes the local attributes and their combinations to form robust global attributes. By establishing connections between local and global features in a self-supervised manner, it adaptively adjusts the granularity of anomaly detection between fine-grained and coarse-grained levels. Joint optimization of these three losses enhances feature discriminability, leading to a more precise and robust anomaly detection performance.

2.3. Fine-tuning with attributes from meta information

In the GLAM fine-tuning stage, we only use the pre-trained teacher encoder and the codebook, adding a linear classifier to predict the clip-level labels. The clip-level labels are manually created from the given meta information, with each unique combination of meta information treated as a new label.

Specifically, for an input audio clip X_w , after extracting the Mel-spectrogram, we perform random masking in both time and frequency dimensions as spec-augmentation [29]. Then, following the method in Sec 2.1, we use the teacher encoder $f_t(\cdot)$ and the codebook C to extract clip-level features $g(x)$, and predict clip-level labels through the linear classifier. We employ ArcFace loss [30] L as the fine-tuning objective, as follows:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cdot (\cos(\theta_{\hat{y}_i} + m))}}{e^{s \cdot (\cos(\theta_{\hat{y}_i} + m))} + \sum_{j=1, j \neq \hat{y}_i}^M e^{s \cdot \cos(\theta_j)}} \quad (8)$$

where s and m are the scaling factor and margin, $\hat{y}_i$ is the label, θ_j is the angle between $g(x)$ and the embedding of the j -th class, with M being the number of classes..

2.4. Anomaly Detection

The pre-trained or fine-tuned model is used to extract clip-level features $g(x)$ and generate memory banks for both the source and target domains for each machine type. For the embedding of a test sample, its anomaly score is calculated as follows:

$$a(\tilde{x}) = \min_i \|g(\tilde{x}) - g(x_i)\|_2 \quad (9)$$

where $\tilde{x}$ is the test sample, and $g(x_i)$ is the corresponding feature in the source or target domain memory bank for its machine type. We then take the maximum of the source and target domain scores as the anomaly score for the test sample.

3. Experiments

3.1. Dataset and Evaluation Metrics

We conducted experiments on the DCASE2023 Task 2 dataset [31, 32], which includes 14 machine types, with domain shifts reflecting different operational states of the machines. The dataset meta information indicates machine type, machine section, and machine attributes. Each machine type has only one machine section, and each section contains 990 samples from the source domain 10 from the target domain. In the evaluation set, each machine section contains 200 normal and anomalous audio samples. Audio clip durations range from 6 to 18 seconds, and are sampled at 16 kHz.

Adhering to the challenge guidelines [3], the area under the receiver operating characteristics curve (AUC) and the partial AUC (pAUC) is used as evaluation metrics for our system. We report the harmonic mean of source AUC (sAUC, %), target AUC (tAUC, %), and pAUC (%) for each machine type, along with the official score Ω (%), which is the harmonic mean of the scores across all machine types.

3.2. Implementation Details

During both the pre-training and fine-tuning stages, only the Development and Additional training data from DCASE2023 Task 2 are used for training, without utilizing any external data.

We extract 128-dimensional Mel-spectrograms as input features, obtained via a 1024-point FFT with a Hamming window of length 1024 and a hop size of 512. Instance normalization [33] is applied to the input features.

During pre-training, we apply data augmentation techniques including white noise injection and fade-in/out effects, with SNRs ranging from 0 to 15 dB. The long clip X^g is set to 6 seconds, while the short clip X^l is set to 2 seconds. The model was trained using the AdamW [34] optimizer with a batch size of 64 over 150 epochs. The learning rate linearly increases to $1e-4$ over the first 10 epochs, and then follows a cosine annealing schedule, decaying to $1e-8$ for the remaining epochs. The Transformer model contains 6 encoder blocks, with 6 attention heads per block and a hidden size of 384. A codebook size of 64 is used, based on ablation experiments presented in Table 3. The temperature parameters for the teacher and student branches, τ_t and τ_s , were set to 0.1 and 0.05, respectively, and λ_1 and λ_2 were set to 1 and 0.1 based on initial tests.

In the fine-tuning stage, we used the AdamW optimizer with a learning rate of $1e-4$, a batch size of 64, and train over 100 epochs. ArcFace loss was used with a scale of 30 and a margin of 0.15.

3.3. Experimental Results

As shown in Table 1, the proposed GLAM method achieves state-of-the-art (SOTA) performance without external data. Despite the use of large-scale pre-trained models in methods like AnoPatch, our approach maintains a significant lead, demonstrating its effectiveness. Additionally, we evaluated GLAM in an unsupervised setting, where only a pre-trained model was used for feature extraction, followed by anomaly detection via KNN. Results confirm that our method performs well even in this scenario.

3.4. Ablation Study

To validate effectiveness, we conducted ablation experiments, with results in Table 2. Systems #1 and #2 employ only L_{LAP}

Table 2: Ablation study for four systems with different settings on the DCASE2023 Task 2 evaluation dataset.

#	Local	Global	sAUC	tAUC	pAUC	Ω
1	LAP	×	72.48	61.13	58.91	63.65
2	×	✓	75.02	71.91	60.42	68.51
3	MSE	✓	78.31	75.23	64.36	72.11
4	LAP	✓	84.06	78.96	68.94	76.79

Table 3: Ablation study for different size of local attributes on DCASE2023 Task 2 evaluation dataset.

Size	sAUC	tAUC	pAUC	Ω
16	81.21	75.90	65.23	73.50
32	84.45	75.04	66.74	74.72
64	84.06	78.96	68.94	76.79
256	82.50	72.21	65.29	72.66
1024	78.77	69.85	63.73	70.25

and L_{KL} in pre-training, respectively, both leading to a significant performance drop, especially for System #1. This highlights the importance of learning global features during pre-training. System #3, following the strategy in [27], uses patch-level mask prediction with MSE loss in the embedding space. The results show that the proposed LAP loss learns more discriminative local features, significantly outperforming the MSE-based approach.

To further analyze performance, we examined the impact of increasing local attribute size from 16 to 1024. Table 3 shows that increasing this size improves performance up to a threshold, beyond which performance degrades. Excessively large local attributes result in sparse global representations from average pooling, weakening the model’s ability to learn relationships between local and global features. Conversely, overly small attributes fail to capture sufficient local information, impairing the audio local distribution model, and reducing performance.

4. Conclusion

This paper has proposed a novel GLAM-ASD method based on global and local attributes mining. A Transformer based encoder containing teacher and student branches is designed to exploit semantic-rich information from patch- and clip-level features in an unsupervised way. The student exploits the context information via masked patch reconstruction, while the teacher is updated from the student via EMA. Local attributes are then learned to describe specific patch-level patterns in machine audio by online clustering, which can also provide supervision for optimizing student through pseudo-labeling. Following DINO, global attributes relate to clip-level patterns, which can be mined by aggregating local attribute mappings. With the help of multi-crop learning, we can exploit the teacher-student architecture again to perform self-distillation without the need for labels. A superior ASD performance of 67.06% and 76.79% in terms of official score Ω has been achieved by multi-granularity attribute learning related to the patch-, clip- and categorical-level representations respectively, demonstrating the effectiveness of GLAM-ASD to address the first-shot problem of the DCASE2023 Task2 challenge.

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6. References

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